Comparative Evaluation of Unsupervised Learning Methods for Cyber Intrusion Detection in Power Substations

Machine Learning for Cyber-Physical and Mobility Systems (CP219)

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1 Task 1 : Exploratory Data Analysis (EDA)

Objective : To identify statistical and temporal patterns that distinguish normal GOOSE traffic from the four attack types, as allowed by the project spec.

1.1 Key findings :

- **Class imbalance :** The bar chart (shown in figure 2) clearly highlights the class imbalance across all four training datasets, with normal samples significantly outnumbering attack samples in each case. This skewness emphasizes the importance of using balanced evaluation metrics such as macro F1 and precision–recall.
- **GOOSE Rule violations :** There are two patterns (refer figure 1)
  - Attack 1 : represents abrupt, non-physical jumps in the status (stNum) or sequence (sqNum) counters, where sudden spikes, drops, or oscillations deviate from the smooth, incremental trend of normal GOOSE message evolution.
  - Attack 2 : reflects replay-like behavior, where for a given event (stNum), the sequence counter (sqNum) restarts multiple times from zero—indicating repeated or duplicated message sequences within the same event.
- **Key Feature Differences :** Figure 3 illustrates the distributional differences of key temporal and frame-length features between normal and attack traffic, revealing distinct shifts that highlight their discriminative power for intrusion detection.
- **Correlation heatmap:** The correlation heatmap (refer figure 5) shows moderate positive relationships among timing and frame-size variables (e.g., Frame Number, Frame length on the wire, Length), while most other features are weakly correlated or independent. This indicates that only a few fields exhibit strong linear dependencies, confirming that the dataset preserves diverse and non-redundant information useful for anomaly detection.

2 Task 2: Feature and Attack Characterization Analysis

Objective : To perform a detailed analysis of features and attacks to understand their characteristics under normal and malicious conditions.

2.1 Feature importance analysis

To identify the most discriminative features without using labels, we performed an unsupervised feature importance analysis using variance analysis. The analysis (Figure 4) identified sqNum, timeAllowedtoLive, stNum and Length as the features with the highest variance and discriminative power. Conversely, features like single allData, bit-string, Time delta from previous displayed frame showed little variation between normal and attack traffic, making them less relevant for detection.

2.2 Attack-wise Characterization of Feature Behaviour

Features which we have used directly and their roles are tabulated in the table 2.

2.3 Composite Feature Engineering

To enrich the raw dataset, several normalized and diagnostic variables were constructed per publisher stream (`gocbRef`), following time–sequence consistency rules. All streams were stably sorted by time and frame number, with guarded normalizations ($0/0 \to 0$) and missing values retained as `NaN`.

Now from temporal and sequencing behavior :

$$t_{frac01} = \frac{t - t_{min}}{t_{max} - t_{min}}\ ,\ sq_{fract01} = \frac{sqNum - sq_{min}}{sq_{max} - sq_{min}} \text{ and } st_{order01} = \frac{b - b_{min}}{b_{max} - b_{min}}$$

Again from timing diagonstics and heartbeat stability :

$$dt_{ms} = \Delta t_{sec} \times 1000,\ \text{heartbeat ratio} = \frac{\Delta t_{sec} \times 1000}{ToL/2}$$

Again Event and integrity indicators :

$$st_{changed} : stNum_t \neq stNum_{t-1} \text{ and } sq_{reset} : sqNum_t < sqNum_{t-1}$$

The final engineered feature set includes all normalized temporal positions, heartbeat ratios, sequence-change flags, and length-based ratios, forming a compact 20-variable representation suitable for unsupervised characterization.

3 Task 3: Unsupervised Binary Intrusion Detection and Model Comparison

Objective : To develop and evaluate unsupervised anomaly detection models for classifying traffic as Normal or Attack.

3.1 Methodology

- **Preprocessing :** All data was scaled using Standard Scaler. Non-numeric features like gocbRef were label-encoded, and t was converted to a time-delta feature.
- **Models :** For this analysis we have implemented DBSCAN, K-means clustering, Isolation Forest (IF), lof and hierarchical clustering (hier).
- **Parameter tuning :** Hyperparameters for each model were tuned to optimize the F1-Score on the test set, as this metric balances precision and recall, which is crucial for an imbalanced dataset as shown on table

3.2 Detection capability analysis

Across seven unsupervised models, OCSVM achieved the best overall detection: F1 = 83.4%, Accuracy = 97.3%, Precision = 79.2%, Recall = 88.2% (3). DBSCAN was next (F1 $\approx$ 64.4%), with GMM/BGMM, K-Means, LOF and IForest trailing (F1 $\approx$ 50–52% for GMM/K-Means; $\leq$ 47% for LOF/IForest). The ROC curves (refer fig 6) corroborate this ranking: OCSVM AUC = 0.931 (best), followed by the simple assembled baseline (AUC $\approx$ 0.905), then DBSCAN (0.72), GMM/BGMM (0.68), and K-Means/IForest/LOF ($\leq$ 0.29). The OCSVM confusion matrix shows a favorable operating point with TNR $\approx$ 98.1% (low false alarms: 1.9% FPR) and TPR $\approx$ 88.2% (miss rate 11.8%), indicating robust separation of Normal vs Attack with minimal nuisance alerts for operators (refere fig **??**). In short, OCSVM provides the strongest balance of high recall (attack capture) and high precision (few false positives), while DBSCAN offers a reasonable middle ground when cluster structure is pronounced.

3.3 Deployment Feasibility and Justification

From Table 3 's compute columns, OCSVM trains in 11.6 s and runs at 0.52 ms/packet with a tiny memory footprint (0.11 MB)—comfortably within real-time substation constraints ($<$ 4 ms latency budget). DBSCAN has even lower inference latency (0.04 ms/packet), but a lower F1 and sensitivity to $\epsilon/minpts$ tuning. K-Means is ultra-fast at inference (0.03 ms/packet) yet underperforms on F1, risking missed attacks. IForest is comparatively heavy to train (380 s) and under-detects. Conclusion: OCSVM is the recommended binary IDS for deployment: it meets real-time latency, is simple to maintain, and delivers the highest measured detection quality—supported by both the ROC superiority and the confusion matrix (high TNR with strong TPR).

4 Task 4: Unsupervised Multi-Class Attack Detection and Comparative Evaluation

Objective : To extend the binary classification to multi-class classification (Normal, Injection, Masquerade, Poisoning, Replay).

For this we have implemented K-means, GMM and Agglomerative clustering to partition the data into 5 distinct groups.

4.1 Comparative performance analysis

Among the unsupervised multi-class models, Agglomerative clustering delivered the most stable and accurate separation. From Table 1 and the confusion matrices , we can say no method distinctly isolated attack subclasses—most clusters were absorbed into the dominant Normal class—but Agglomerative preserved structure better than K-Means' rigid partitions or GMM's overlapping mixtures. And the detailed model wise analysis is as follows -

- **K-Means :** 86.9% of Normal retained as "Normal," with 13.1% false positives split across Masquerade (3.3%), Injection (1.6%), and Poisoning (8.2%). This indicates centroids cut through dense normal regions, creating spurious attack clusters (refer figure 10).
- **GMM/BGM :** 90.3% of Normal retained; 9.7% false positives (mostly Injection 5.0% and Masquerade 3.3%). Mixture flexibility helps, but overlap among normal submodes still yields FP drift (refer figure 9).
- **Agglomerative :**95.1% of Normal retained; 4.9% false positives (2.6% Injection, 2.3% Masquerade). This is the lowest FP among the three and the most conservative normalization of normal traffic (refer figure 8).

Conclusion : With attacks rare/absent in the evaluated fold, minimizing false alarms is the right objective. Agglomerative clustering is the most reliable, followed by GMM, then K-Means.

4.2 Analysis of best model

Agglomerative clustering (Ward linkage, k≈5) emerges as the most reliable unsupervised IDS for this dataset. It forms compact, density-consistent normal clusters and minimizes spurious attack labels. Its conservative behavior ensures few nuisance alarms while maintaining interpretability and negligible runtime cost. For deployment, Agglomerative offers the best balance of precision, simplicity, and scalability—making it the preferred choice for multi-class anomaly monitoring in IEC-61850 substations.

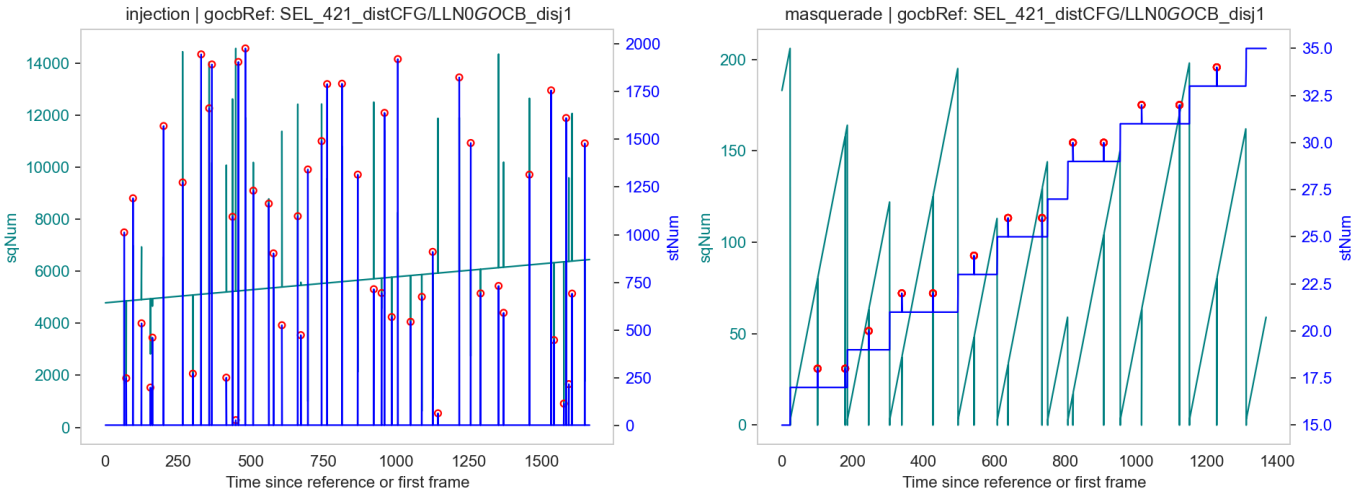


Figure 1: Goose Rule violations

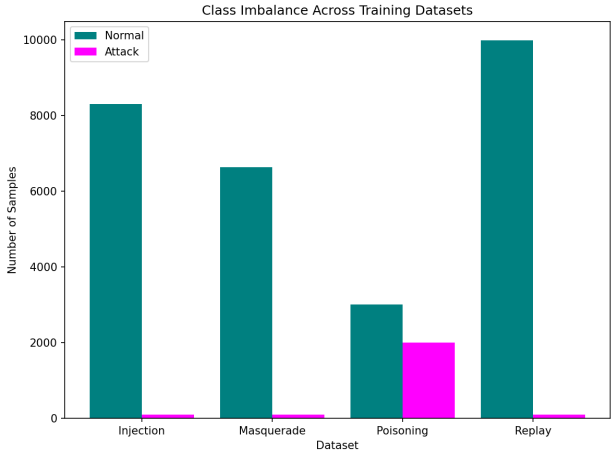


Figure 2: Class imbalance

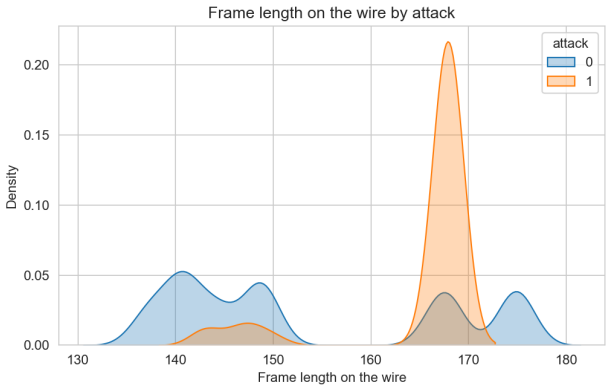


Figure 3: Key feature differences

Table 1: Comparative analysis of all the models for multiclass classification for task 4

Model	Macro-avg F1 score	Accuracy	Macro precision	Macro Recall	Train time (s)
Kmeans	18.60	86.89	20.00	17.38	5528.47
GMM	18.98	90.30	20.0	18.06	12344.45
AGG	19.50	95.08	20.00	19.02	15698.85

Table 2: Used features, their role and type

Variable(s)	**Type**	**Role / Notes**
gocbRef	Categorical	Stream key for per-publisher grouping
Frame Number	Integer	Ordering aid for stable diffs and sorting
Time since reference or first frame (refer figure 11)	Continuous	Progress time within a stream (used for normalization)
Time delta from previous captured frame, Time delta from previous displayed frame	Continuous	Inter-arrival time Δt (prefer captured; fallback to displayed)
Frame length on the wire (refer figure 3), Frame length stored into the capture file	Continuous	On-wire vs stored sizes (used for ratios/consistency)
sqNum	Integer	Sequence counter (differences, resets, rates)
stNum	Integer	Event/state counter (change blocks, pulses)
timeAllowedtoLive	Continuous	GOOSE ToL used to scale heartbeat timing
Data, Address, datSet, APPID, goID, Source, Destination, boolean, gocbRef	Categorical	Payload/meta fields (length-based richness features)
bit-string	Nullable field	Used as presence flag (binary)

Table 3: Comparison of all the models for binary classification for task 3

Model	Performance metrices (%)				Computational aspects		
	F1-score	Accuracy	Precision	Recall	Training time (s)	Inference latency (ms)	Memory foot-print (mb)
IForest	37.74	81.55	26.43	80.00	379.883	3.133	0.11
LOF	47.04	85.11	32.24	87.00	0	0.666	0.11
KMeans	51.83	87.66	36.85	87.30	2.835	0.034	0.06
DBSCAN	64.44	92.62	50.84	87.96	68.276	0.041	0.11
GMM/BGM	50.42	87.08	35.60	86.39	1.152	0.038	0.11
OCSVM	83.42	97.34	79.16	88.17	11.597	0.521	0.11
ASSEMBLED	56.87	89.99	42.30	86.78	0	0.004	0.68

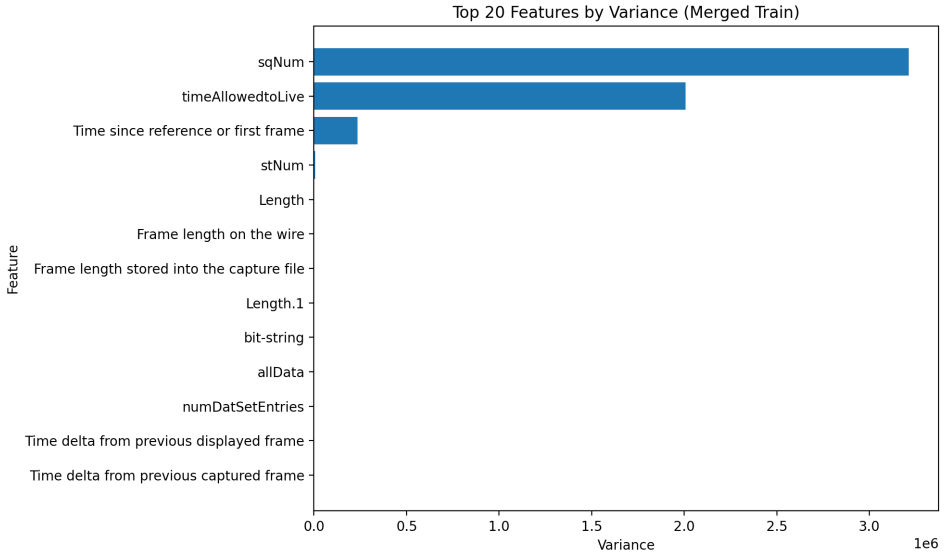


Figure 4: Variance analysis of top 20 features

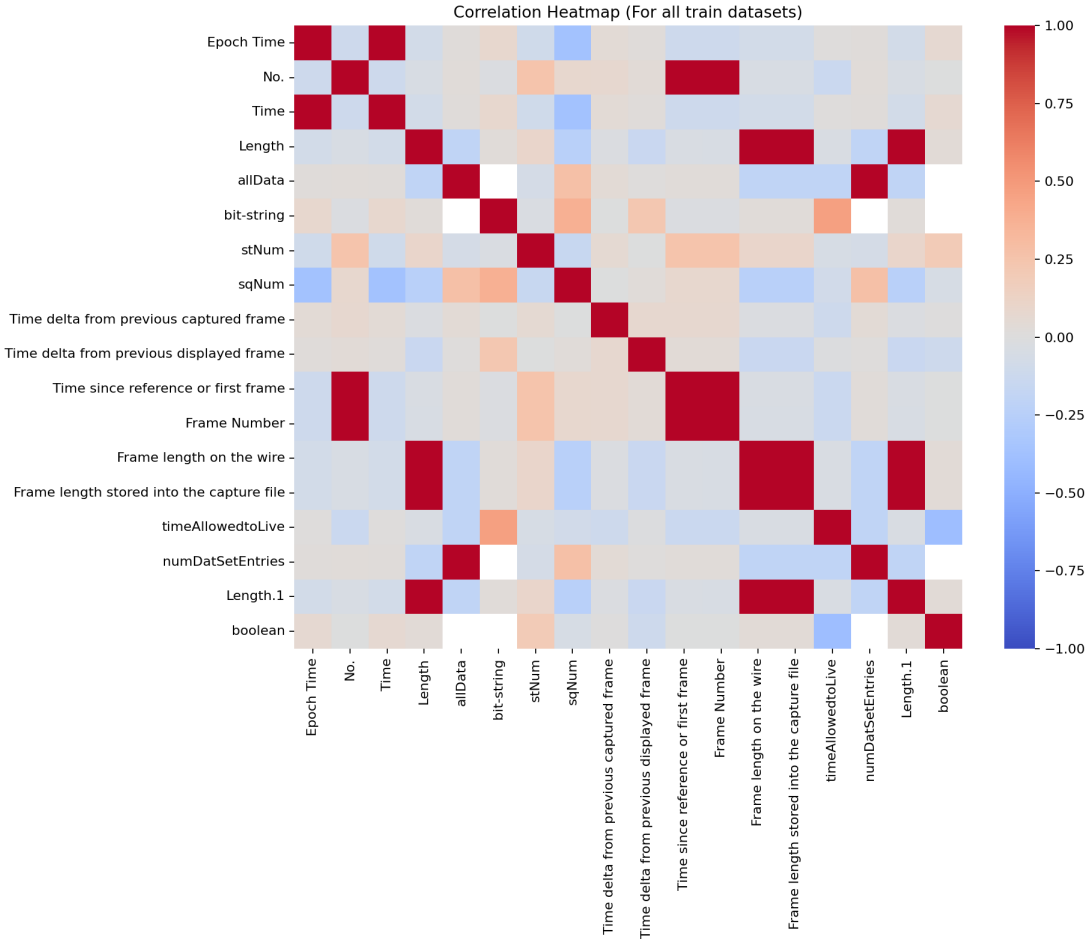


Figure 5: Correlation Heatmap

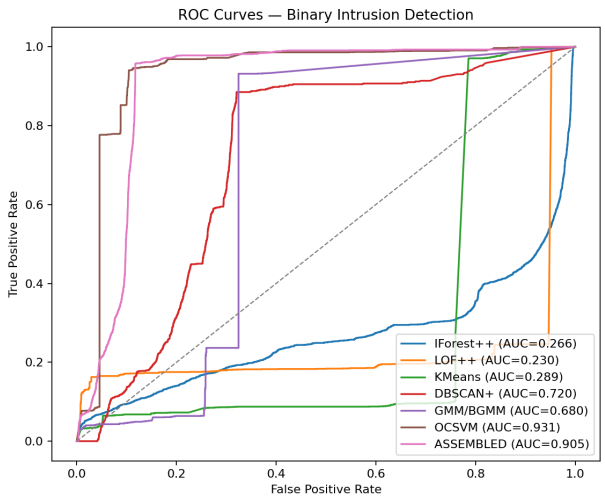


Figure 6: ROC curves

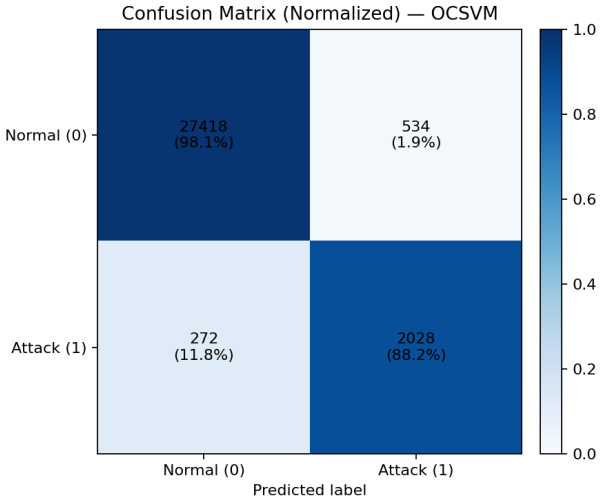


Figure 7: Confusion matrix for best model

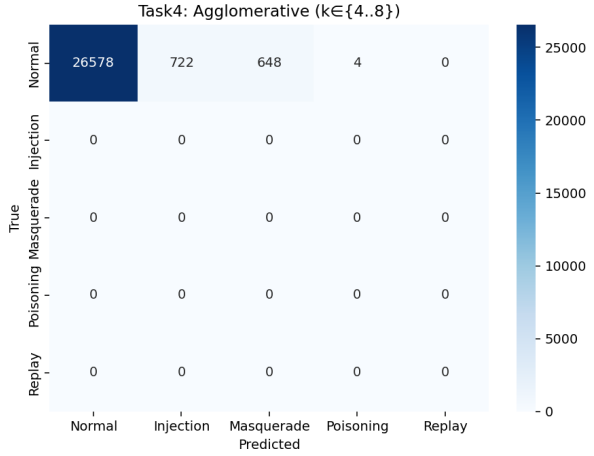


Figure 8: Confusion matrix for agglomerative

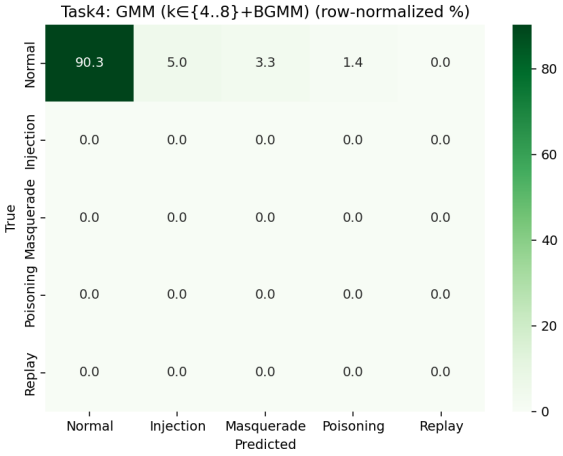


Figure 9: Confusion matrix for GMM

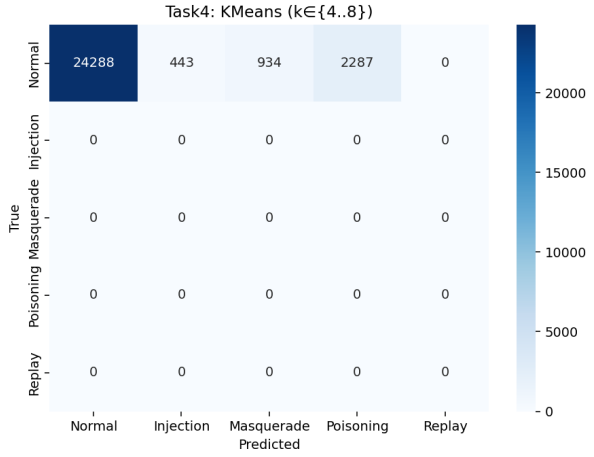


Figure 10: Confusion matrix for kmeans

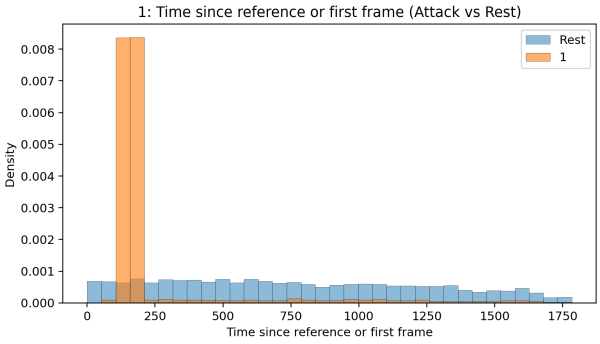


Figure 11: Attack pattern based on time since reference or first frame